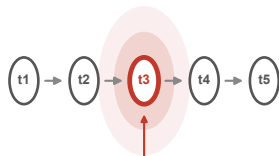


a

Which step is fragile?

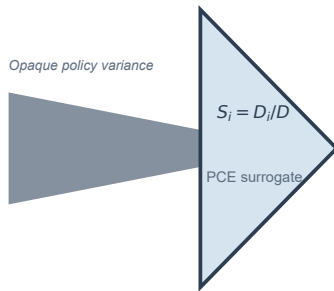


Uncertainty spikes here

Reward uncertainty propagates non-uniformly across steps

b

Opaque policy variance



1. Parameterise reward via PCA

2. Train ensemble (L = 20-50)

3. Fit PCE surrogate

4. Extract Sobol analytically

c

Interpretable Sobol indices

S_1 — PC1 (reward mean)
Catalyst

S_2 — PC2 (reward spread)
Base

S_3 — residual
Aryl halide

